



RIVER VALLEY HIGH SCHOOL

JC 2 PRELIMINARY EXAMINATIONS

H3 BIOLOGY

9816/01

Paper 1

24 September 2021

INSERT

2 hours 30 minutes

INSTRUCTIONS

You may annotate this insert and use any blank spaces for planning. **Do not write your answers** on the insert.

Information for Question 1

The information provided in this insert is based on a number of published scientific articles. Other published articles may not agree with all of this information.

Endocytosis and eukaryote evolution

Endosymbiosis was a key breakthrough early in cell history. Only cells that possessed mitochondria had the energetic means to attain eukaryotic cell complexity, which include multicellularity.

According to the endosymbiotic theory, the genomes of mitochondria are derived from engulfed ancestral bacteria of the α -proteobacteria lineage. However, current mitochondrial genome sizes are much smaller than those of descendant free-living bacteria from the ancestral bacteria. One process that led to reduced mitochondrial DNA (mtDNA) genome size is genetic transfer from mitochondria to the nucleus.

Some studies researching the mode of genetic transfer found that:

- Genes in mtDNA of different lineages have been transferred to the nuclear genome multiple times in evolutionary history before being lost from the mitochondria genome.
- A single-subunit RNA polymerase is present in mitochondria.
- In many species, introns that are present in the mitochondrial genes in related species are absent in their nuclear copies of these same mitochondrial genes.

Different genes remain in the mitochondria in species of different ancestral lineages. For example, in vertebrates such as humans, the mtDNA contains 37 genes that encode for 22 tRNAs, 2 rRNAs and 13 proteins involved in the oxidative phosphorylation system. In the mitochondria of invertebrates, the mtDNA also contains genes encoding for ribosomal proteins.

Evolution of multicellularity in animals

All multicellular organisms evolved from unicellular ancestors. First known single-celled organisms appeared on Earth about 3.5 billion years ago before more complex forms appeared about 600 million years ago.

By studying the genomic and biological features of extant¹ unicellular and multicellular Animalia species, coupled with analyses based on a well-supported phylogenetic framework, researchers reconstructed the last unicellular ancestor (LUA) and the last common multicellular ancestor (LCMA) of animals. LUA represents the single-celled ancestor immediately preceding the emergence of the multicellular animal, and LCMA represents the last common ancestor that gave rise to all living animals today.

¹ Currently still in existence.

Fig. 1.1 shows the reconstructed last unicellular ancestor (LUA).

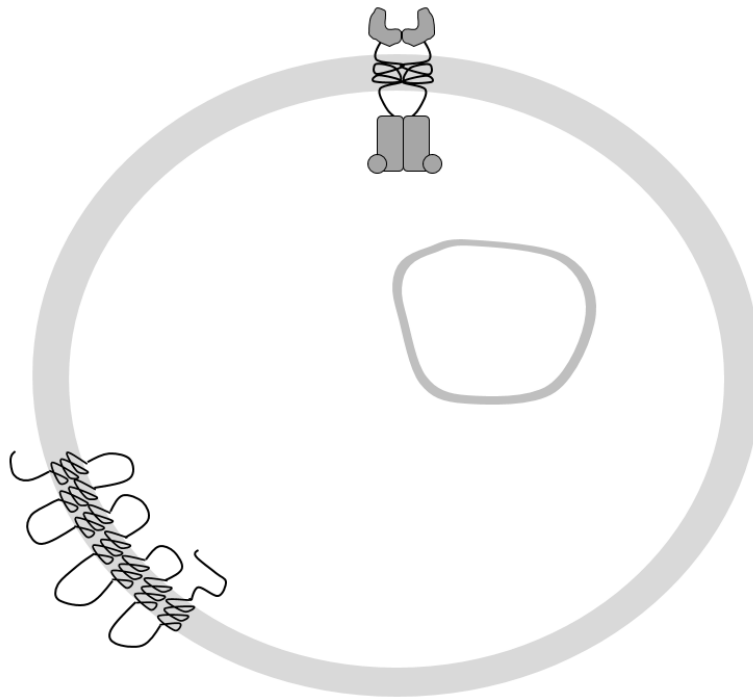


Fig. 1.1

Fig. 1.2 shows two adjacent cells of the reconstructed last common multicellular ancestor (LCMA).

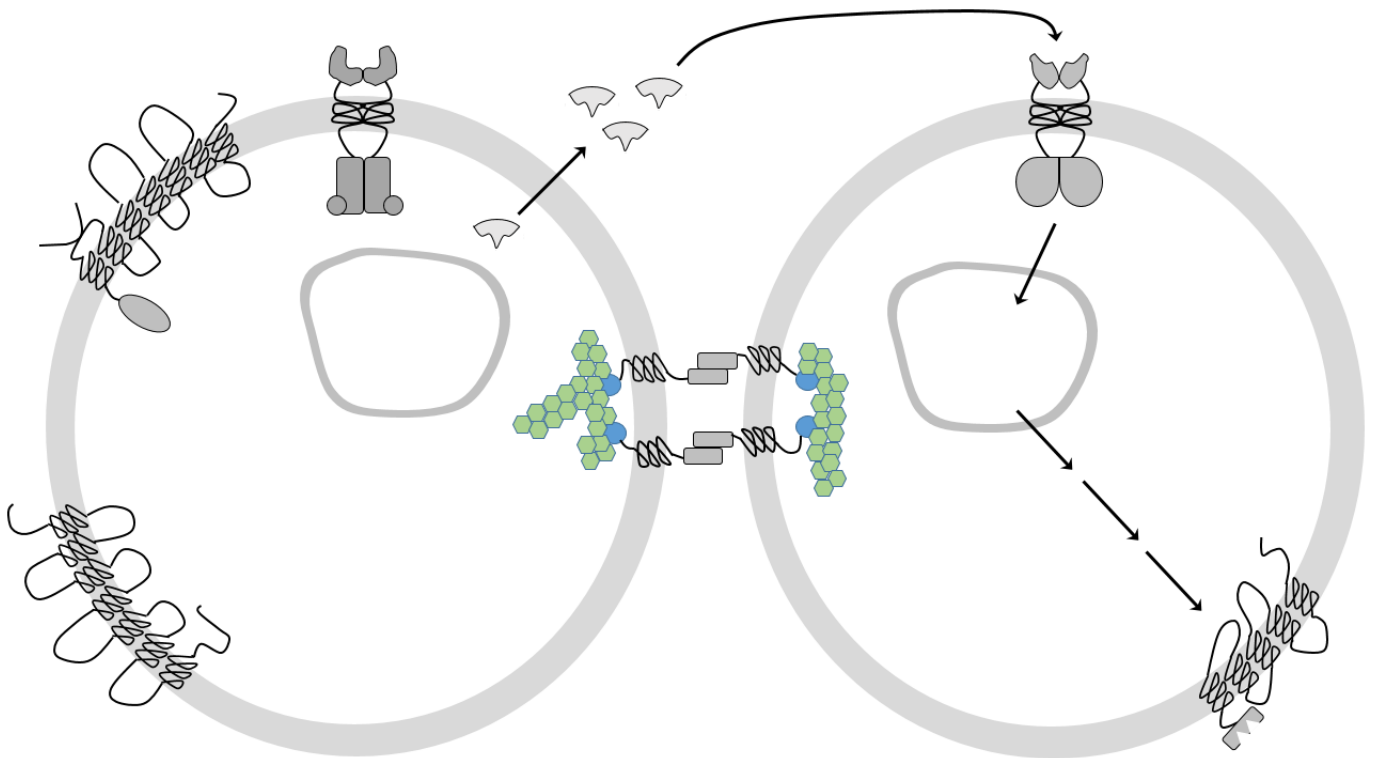


Fig. 1.2

Researchers proposed that LUA possessed a life cycle comprising a sexually reproductive stage made of individual cells and at least one multicellular stage. In LUA, the multicellular stage consists of an assembly of cells that are capable of responding individually to environmental stimuli instead of a coordinated response with the other cells. In LCMA, cells in the multicellular stage were further able to integrate information within the structure to result in a coordinated response and also an organisation of cells types. This later evolved to result in the formation of germ layers (endoderm, mesoderm, ectoderm) during embryonic development which allowed the formation of complex body plans seen in modern animals.

The evolution of LUA genome gave rise to molecular complexity in LCMA. This molecular complexity allowed cells in the different regions of LCMA to respond differently at the same time, therefore resulting in the ability to integrate information within the structure.

Table 1.1 shows the presence or absence of selected genes in the LUA and LCMA genome.

Table 1.1

gene coding for	LUA	LCMA
molecule A	present	present
molecule A variant 1	absent	present
molecule A variant 2	absent	present
molecular target of A	present	present
molecule B	present	present
molecular target of B	absent	present
molecule C	absent	present
molecular target of C	absent	present

Fig. 1.3 shows the regions 1, 2 and 3 referred to in Fig. 1.4 for the multicellular stage of both LUA and LCMA.

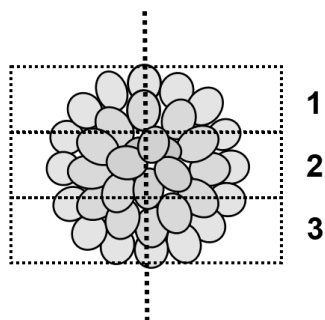


Fig. 1.3

Fig. 1.4 shows the distribution of expressed molecules, from Table 1.1, present in the different regions of the LUA and LCMA multicellular stage at different hours after its formation.

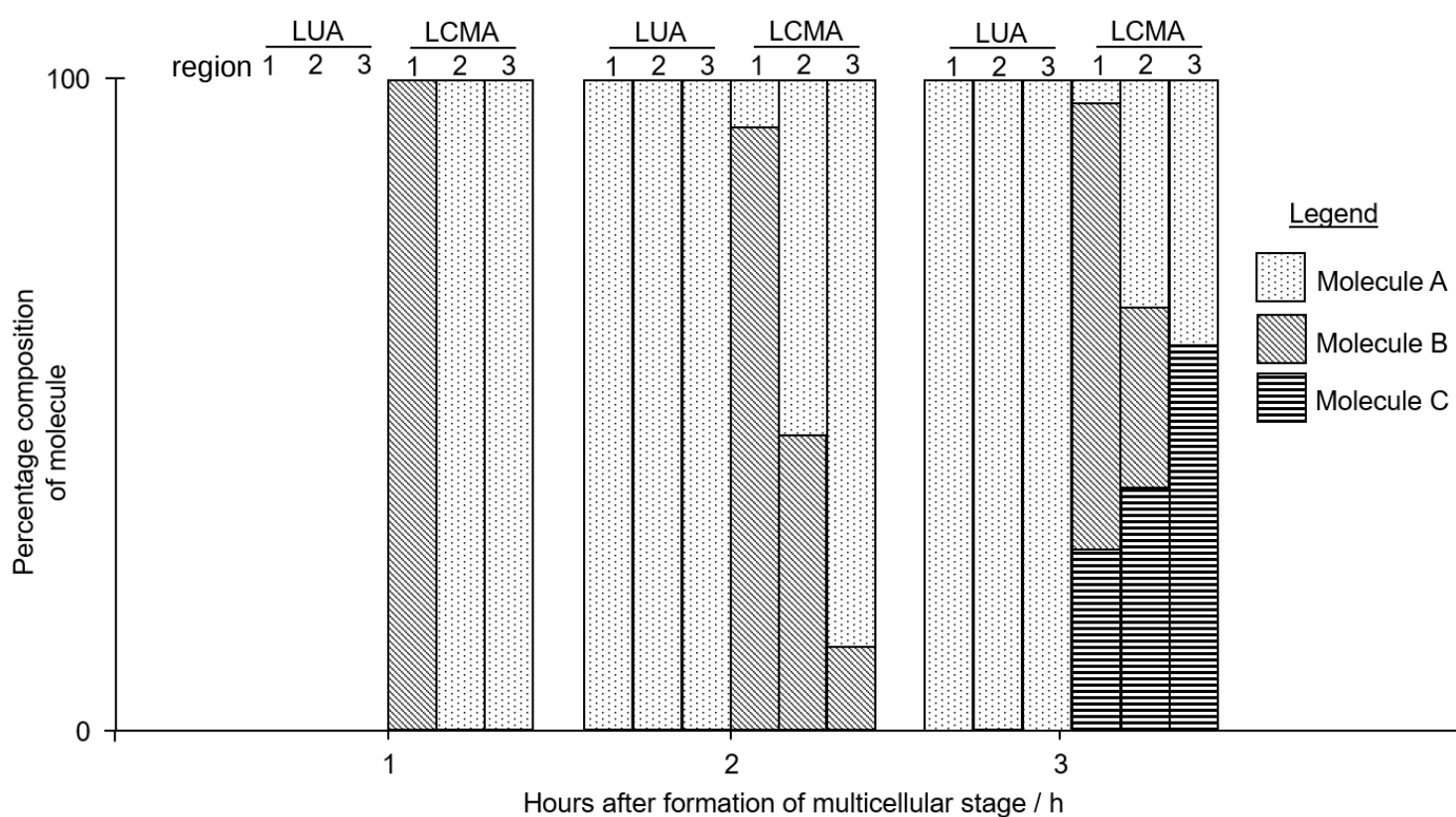


Fig. 1.4

Mutation and eukaryote evolution

Mutations generate genetic variation for other evolutionary forces to operate on. Mutations in mtDNA are particularly important due to the roles mitochondria have in cells and also the compact nature of mtDNA, which means most of the sequences are coding. However, unlike diploid nuclear DNA, mtDNA exists in a highly polyploidy state within each cell. mtDNA is hyper-mutable due to damage caused by exposure to highly reactive chemicals such as reactive oxygen species² as well as being replicated more frequently.

An investigation was made to identify the mutation frequency in different sequences of mtDNA and its relationship to both cell type and age of multicellular organisms. Mice belonging to two age groups, 20 day-old pups and 10 month-old mothers, were analysed.

Fig. 1.5 shows the results of this investigation. The D-loop in mtDNA is a non-coding sequence which regulates mtDNA replication and transcription. All differences between pups and mothers have a p-value < 0.05.

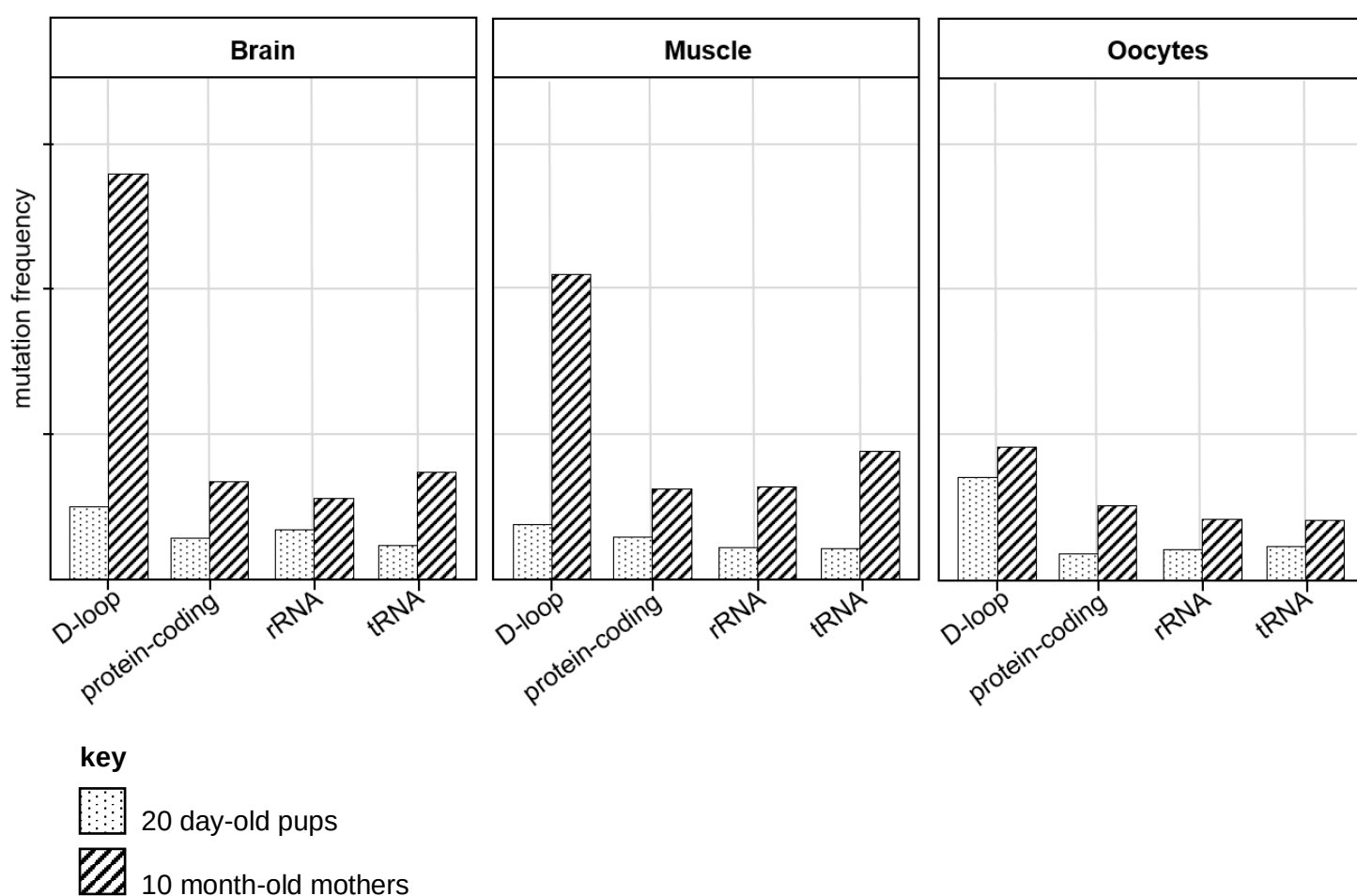


Fig. 1.5

² Highly reactive chemicals formed from oxygen, including peroxides and superoxide.

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